

创建 bash 文件并且修改权限使其能够被执行

vi menu.sh	# 创建 bash 文件
./menu.sh	# 尝试运行文件并且发现没有权限
man chmod	# 查看 chmod 使用方法
ls -l	# 确认文件权限
chmod +x menu.sh	# 修改文件权限，增加可执行权限
ls -l	# 确认文件权限
./menu.sh	# 执行 menu.sh

```
zhangmi@raspberrypi:~/Desktop/HW1 $ vi menu.sh
zhangmi@raspberrypi:~/Desktop/HW1 $ ./menu.sh
-bash: ./menu.sh: Permission denied
zhangmi@raspberrypi:~/Desktop/HW1 $ man chmod
zhangmi@raspberrypi:~/Desktop/HW1 $ ls -l
total 4
-rw-r--r-- 1 zhangmi zhangmi 1618 Nov  9 07:55 menu.sh
zhangmi@raspberrypi:~/Desktop/HW1 $ chmod +x menu.sh
zhangmi@raspberrypi:~/Desktop/HW1 $ ls -l
total 4
-rwxr-xr-x 1 zhangmi zhangmi 1618 Nov  9 07:55 menu.sh
zhangmi@raspberrypi:~/Desktop/HW1 $ ./menu.sh
Please select an option:
a. List all files in the present working directory
b. Display today's date and time
c. Display whether a file is a 'simple' file or a 'directory'
d. Create a backup for a file using 'tar' command
e. Compress the tar file
f. Start an ftp session
g. Start your LED control program
x. Exit
Enter your choice: a
```

menu.sh 代码

```
#!/bin/bash
```

```
while true; do
    echo "Please select an option:"
    echo "a. List all files in the present working directory"
    echo "b. Display today's date and time"
    echo "c. Display whether a file is a 'simple' file or a 'directory'"
    echo "d. Create a backup for a file using 'tar' command"
    echo "e. Compress the tar file"
    echo "f. Start an ftp session"
    echo "g. Start your LED control program"
    echo "x. Exit"
    read -p "Enter your choice: " option
```

```

case $option in
a)
    ls -l
    ;;
b)
    date
    ;;
c)
    read -p "Enter the file name: " filename
    if [ -f "$filename" ]; then
        echo "It's a file."
    elif [ -d "$filename" ]; then
        echo "It's a directory."
    else
        echo "File not found."
    fi
    ;;
d)
    read -p "Enter the file name to backup: " filename
    read -p "Enter the backup file name: " backupname
    tar -cvf $backupname $filename
    ;;
e)
    read -p "Enter the tar file name to compress: " tarfilename
    gzip $tarfilename
    ;;
f)
    read -p "Enter the domain name (or IP address): " domain
    ftp $domain
    ;;
g)
    if [ -x "/home/zhangmi/Desktop/gpio-led.py" ]; then
        python /home/zhangmi/Desktop/gpio-led.py
    else
        echo "LED control program not found."
    fi
    ;;
x)
    echo "Exiting..."
    exit 0
    ;;
*)
    echo "Invalid option, please try again."
    ;;

```

```
esac
done
```

在树莓派上安装 ftp

```
Please select an option:
a. List all files in the present working directory
b. Display today's date and time
c. Display whether a file is a 'simple' file or a 'directory'
d. Create a backup for a file using 'tar' command
e. Compress the tar file
f. Start an ftp session
g. Start your LED control program
x. Exit
Enter your choice: f
Enter the domain name (or IP address): 10.32.134.152
./menu.sh: line 43: ftp: command not found
```

```
zhangmi@raspberrypi:~ $ sudo apt-get update
Hit:1 http://archive.raspberrypi.com/debian bookworm InRelease
Hit:2 http://raspbian.raspberrypi.com/raspbian bookworm InRelease
Reading package lists... Done
W: http://raspbian.raspberrypi.com/raspbian/dists/bookworm/InRelease: Key is stored in legacy trusted.gpg keyring (/etc/apt/trusted.gpg), see the DEPRECATION section in apt-key(8) for details.
zhangmi@raspberrypi:~ $ sudo apt-get install ftp
```

在一段等待之后进入了 ftp 服务

```
zhangmi@raspberrypi:~/Desktop/HW1 $ ./menu.sh
Please select an option:
a. List all files in the present working directory
b. Display today's date and time
c. Display whether a file is a 'simple' file or a 'directory'
d. Create a backup for a file using 'tar' command
e. Compress the tar file
f. Start an ftp session
g. Start your LED control program
x. Exit
Enter your choice: f
Enter the domain name (or IP address): 10.32.134.152
ftp: Can't connect to `10.32.134.152:21': Connection timed out
ftp: Can't connect to `10.32.134.152:ftp'
ftp> █
```

解决 g.Start your LED control program 一直显示 LED control program not found 的问题

期望执行的文件为 /home/zhangmi/Desktop/gpio-led.py，一开始发现一直显示 LED control program not found，检查文件权限之后发现问题：gpio-led.py 没有执行权限。运行 `chmod +x gpio-led.py` 之后解决问题。

```
zhangmi@raspberrypi:~/Desktop/HW1 $ ./menu.sh
Please select an option:
a. List all files in the present working directory
b. Display today's date and time
c. Display whether a file is a 'simple' file or a 'directory'
d. Create a backup for a file using 'tar' command
e. Compress the tar file
f. Start an ftp session
g. Start your LED control program
x. Exit
Enter your choice: g
LED control program not found.
Please select an option:
a. List all files in the present working directory
b. Display today's date and time
c. Display whether a file is a 'simple' file or a 'directory'
d. Create a backup for a file using 'tar' command
e. Compress the tar file
f. Start an ftp session
g. Start your LED control program
x. Exit
Enter your choice: x
Exiting...
zhangmi@raspberrypi:~/Desktop/HW1 $ cd ../
zhangmi@raspberrypi:~/Desktop $ ls -l
total 44
drwxr-xr-x 2 zhangmi zhangmi 4096 Oct  7 01:51 'gpio led_c'
-rw-r--r-- 1 zhangmi zhangmi  527 Nov  9 23:26 gpio-led.py
drwxr-xr-x 2 zhangmi zhangmi 4096 Oct  7 01:50 helloworld_c
drwxr-xr-x 3 zhangmi zhangmi 4096 Nov  9 23:42 HW1
drwxr-xr-x 2 zhangmi zhangmi 4096 Oct 16 04:18 lab3
drwxr-xr-x 2 zhangmi zhangmi 4096 Oct 16 04:19 lab4
drwxr-xr-x 2 zhangmi zhangmi 4096 Oct 23 05:16 lab5
drwxr-xr-x 2 zhangmi zhangmi 4096 Nov  1 22:00 lab6
drwxr-xr-x 2 zhangmi zhangmi 4096 Nov  9 07:23 lab7
drwxr-xr-x 2 zhangmi zhangmi 4096 Sep 18 06:09 __pycache__
-rw-r--r-- 1 zhangmi zhangmi   36 Oct  7 08:52 'wifi password'
```



```

zhangmi@raspberrypi:~/Desktop $ chmod +x gpio-led.py
zhangmi@raspberrypi:~/Desktop $ ls -l
total 44
drwxr-xr-x 2 zhangmi zhangmi 4096 Oct  7 01:51 'gpio_led_c'
-rwxr-xr-x 1 zhangmi zhangmi  527 Nov  9 23:26 gpio-led.py
drwxr-xr-x 2 zhangmi zhangmi 4096 Oct  7 01:50 helloworld_c
drwxr-xr-x 3 zhangmi zhangmi 4096 Nov  9 23:42 HW1
drwxr-xr-x 2 zhangmi zhangmi 4096 Oct 16 04:18 lab3
drwxr-xr-x 2 zhangmi zhangmi 4096 Oct 16 04:19 lab4
drwxr-xr-x 2 zhangmi zhangmi 4096 Oct 23 05:16 lab5
drwxr-xr-x 2 zhangmi zhangmi 4096 Nov  1 22:00 lab6
drwxr-xr-x 2 zhangmi zhangmi 4096 Nov  9 07:23 lab7
drwxr-xr-x 2 zhangmi zhangmi 4096 Sep 18 06:09 __pycache__
-rw-r--r-- 1 zhangmi zhangmi   36 Oct  7 08:52 'wifi password'

```

拓展：增加新功能 CPU and Memory Usage 以及 Network Speed

Test

```

Please select an option:
a. List all files in the present working directory
b. Display today's date and time
c. Display whether a file is a 'simple' file or a 'directory'
d. Create a backup for a file using 'tar' command
e. Compress the tar file
f. Start an ftp session
g. Start your LED control program
h. System Monitoring
    i. CPU and Memory Usage
    ii. Network Speed Test
x. Exit
Enter your choice: h

```

menu-upgraded.sh 代码

```
#!/bin/bash
```

```

while true; do
    echo "Please select an option:"
    echo "a. List all files in the present working directory"
    echo "b. Display today's date and time"
    echo "c. Display whether a file is a 'simple' file or a 'directory'"
    echo "d. Create a backup for a file using 'tar' command"
    echo "e. Compress the tar file"
    echo "f. Start an ftp session"
    echo "g. Start your LED control program"
    echo "h. System Monitoring"
    echo "    i. CPU and Memory Usage"

```

```

echo "    ii. Network Speed Test"
echo "x. Exit"
read -p "Enter your choice: " option

case $option in
    a)
        ls -l
        ;;
    b)
        date
        ;;
    c)
        read -p "Enter the file name: " filename
        if [ -f "$filename" ]; then
            echo "It's a file."
        elif [ -d "$filename" ]; then
            echo "It's a directory."
        else
            echo "File not found."
        fi
        ;;
    d)
        read -p "Enter the file name to backup: " filename
        read -p "Enter the backup file name: " backupname
        tar -cvf $backupname $filename
        ;;
    e)
        read -p "Enter the tar file name to compress: " tarfilename
        gzip $tarfilename
        ;;
    f)
        read -p "Enter the domain name (or IP address): " domain
        ftp $domain
        ;;
    g)
        if [ -x "/home/zhangmi/Desktop/gpio-led.py" ]; then
            python /home/zhangmi/Desktop/gpio-led.py
        else
            echo "LED control program not found."
        fi
        ;;
    h)
        echo "Select a monitoring option:"
        echo "i. CPU and Memory Usage"

```

```

echo "ii. Network Speed Test"
read -p "Enter your choice: " monitor_option
case $monitor_option in
    i)
        # CPU and Memory Usage
        if command -v htop &> /dev/null; then
            htop
        else
            top
        fi
        ;;
    ii)
        # Network Speed Test
        speedtest-cli
        ;;
    *)
        echo "Invalid option, please try again."
        ;;
esac
;;
x)
echo "Exiting..."
exit 0
;;
*)
echo "Invalid option, please try again."
;;
esac
done

```

Install speedtest-cli 以实现网络测速功能

```

zhangmi@raspberrypi:~ $ sudo apt-get install speedtest-cli
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following NEW packages will be installed:
  speedtest-cli
0 upgraded, 1 newly installed, 0 to remove and 285 not upgraded.
Need to get 24.6 kB of archives.
After this operation, 106 kB of additional disk space will be used.
Get:1 http://mirrors.bfsu.edu.cn/raspbian/raspbian bookworm/main armhf speedtest
-cli all 2.1.3-2 [24.6 kB]
Fetched 24.6 kB in 1s (17.7 kB/s)
Selecting previously unselected package speedtest-cli.
(Reading database ... 213959 files and directories currently installed.)
Preparing to unpack .../speedtest-cli_2.1.3-2_all.deb ...
Unpacking speedtest-cli (2.1.3-2) ...
Setting up speedtest-cli (2.1.3-2) ...
Processing triggers for man-db (2.11.2-2) ...
zhangmi@raspberrypi:~ $ speedtest-cli
Retrieving speedtest.net configuration...
Testing from China Education and Research Network Center (110.65.147.237)...
Retrieving speedtest.net server list...
Selecting best server based on ping...
Hosted by Netvigator (Hong Kong) [1389.09 km]: 107.879 ms
Testing download speed.....
Download: 12.31 Mbit/s
Testing upload speed.....
Upload: 38.60 Mbit/s

```